

Metrics

1. Core counters (state / totals)

- `step_count`
- `prompt_tokens_total`
- `generation_tokens_total`

2. Latency tracking (raw samples)

These are stored as lists of per-request or per-event measurements:

- `prefill_latencies` (seconds per prefill phase)
- `decode_latencies` (seconds per decode phase)
- `e2e_latencies` (end-to-end request latency, seconds)
- `ttft_latencies` (time-to-first-token, seconds)
- `inter_token_deltas` (time between generated tokens, seconds)

3. Input statistics

- `prompt_lengths` (token counts per prompt/request)

4. Internal tracking (not exposed directly as metrics)

Used for timing but not exported directly:

- `_request_start` (group_id → start time)
- `_request_first_token` (group_id → first token time)
- `_last_token_time` (group_id → last token time)

5. Step-level runtime metrics (printed in `print_step`)

These are computed per step:

- `prefill_ms`
- `decode_ms`
- `gen_tokens`
- `tok/s` (tokens per second, computed as `gen_tokens / (prefill+decode time)`)

- `running`
- `waiting`
- `swapped`
- `block_util` (KV cache utilization ratio)

Derived display helpers:

- `_kv_bar(block_util)` (visual representation only)

6. System metrics (optional, runtime-dependent)

GPU (if torch available)

- `gpu_mem` = allocated / reserved memory (MB)

CPU (if psutil available)

- `cpu` = CPU usage percentage

7. Summary metrics (`print_summary`, `to_csv_rows`)

These are aggregated statistics:

Latency averages

- `time_to_first_token_avg_ms`
- `tpot_avg_ms` (time per output token)
- `e2e_request_latency_avg_ms`
- `prefill_latency_avg_ms`
- `decode_latency_avg_ms`

Throughput / volume

- `prompt_tokens_total`
- `generation_tokens_total`

Input stats

- `prompt_len_avg`

8. CSV export fields (**save_csv**)

Final exported schema:

- `prompt_tokens_total`
- `generation_tokens_total`
- `time_to_first_token_avg_ms`
- `tpot_avg_ms`
- `e2e_request_latency_avg_ms`
- `prompt_len_avg`
- `prefill_latency_avg_ms`
- `decode_latency_avg_ms`
- `experiment`
- `timestamp`